

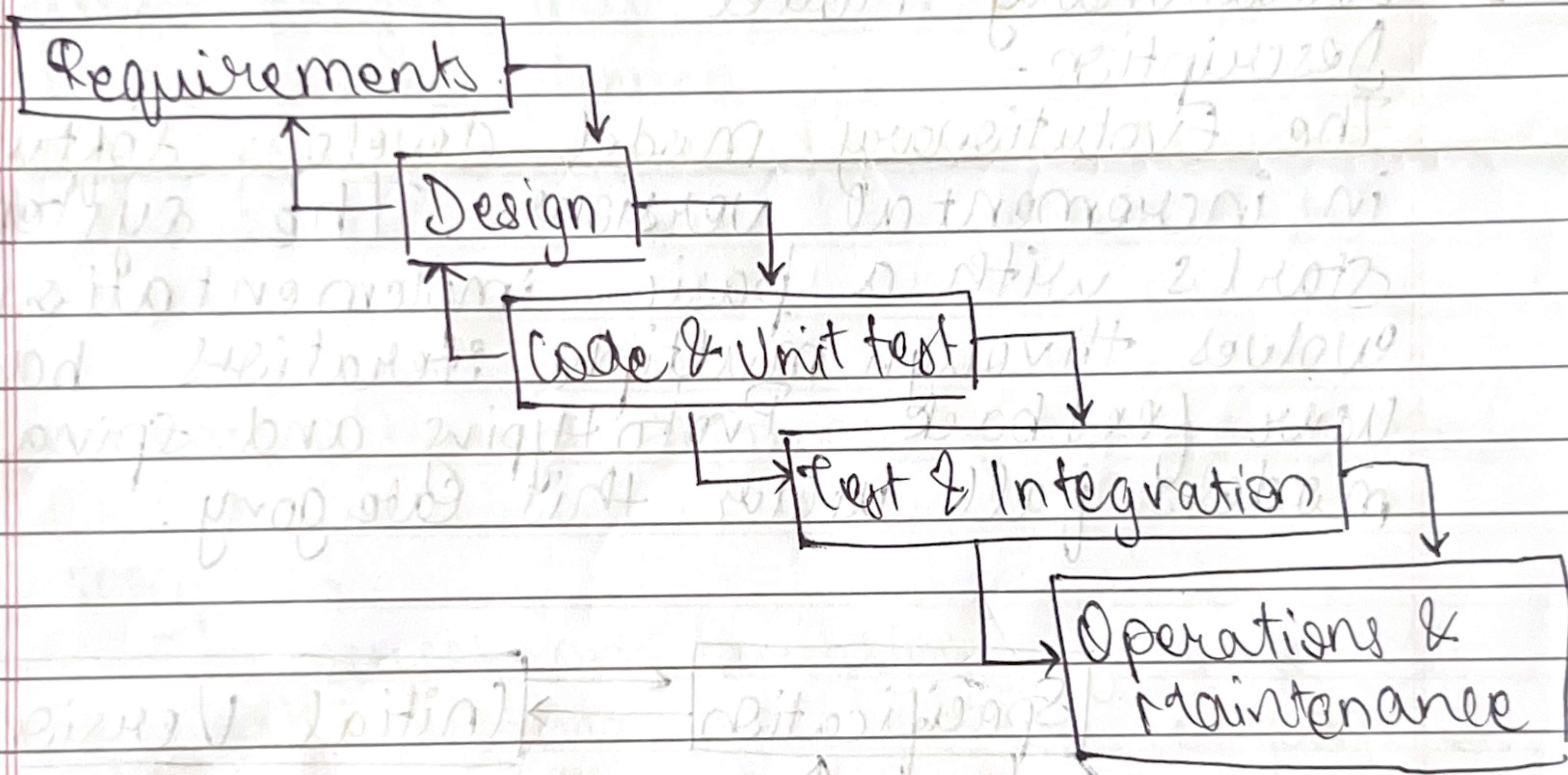
18/01/26

## Software Process Models.

### 1. Waterfall Model

#### Description -

The Waterfall model is a linear and sequential software development process. Each phase must be completed before the next one begins. The phases typically include Requirements Analysis, Design, Implementation, Testing, Development and Maintenance. There is no overlap between phases.



#### Advantages

- Easy to understand and implement
- Clearly defined stages and documentation
- Simple to manage due to its structured nature
- Suitable for small projects with stable requirements.

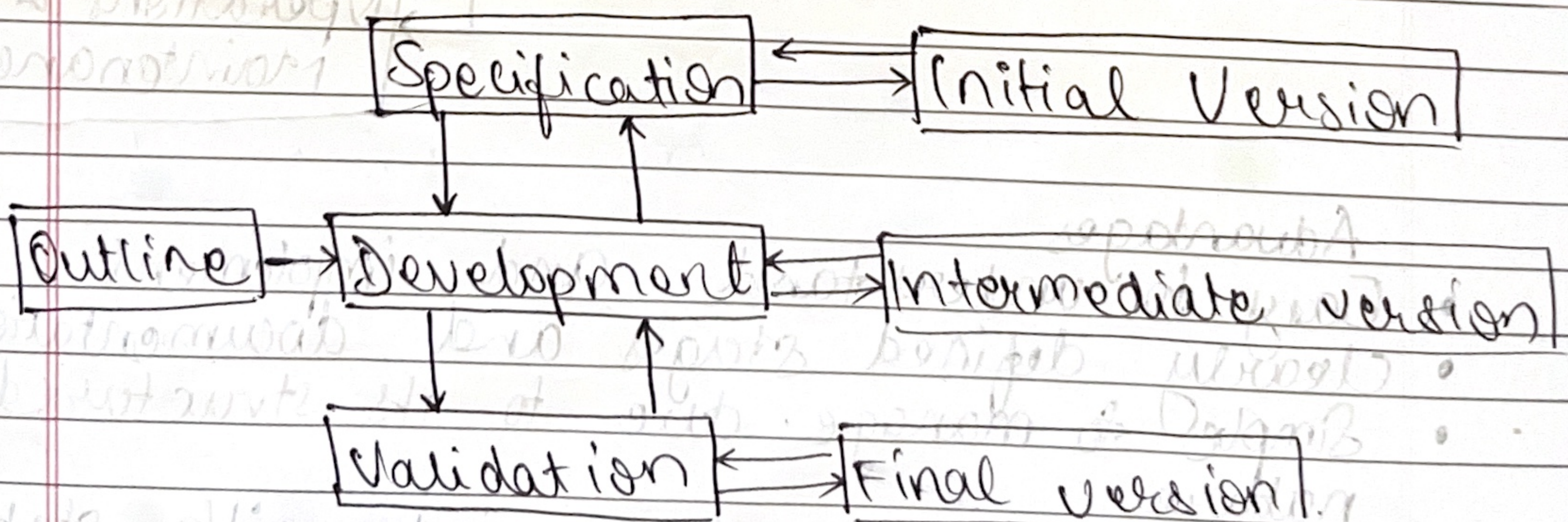
## Disadvantages.

- Very rigid; changes are difficult after a phase is completed.
- Working software is available only at the end.
- High risk if requirements are incorrect or misunderstood.
- Not suitable for complex or long duration projects.

## 2 Evolutionary Model

### Description -

The Evolutionary model develops software in incremental versions. The system starts with a basic implementation and evolves through multiple iterations based on user feedback. Prototyping and spiral models fall under this category.



## Advantages

- Early delivery of partial working software.
- Continuous user feedback improves requirement clarity.
- Flexible and adapts well to changing requirements.
- Reduces technical and requirements-related risks.

## Disadvantages

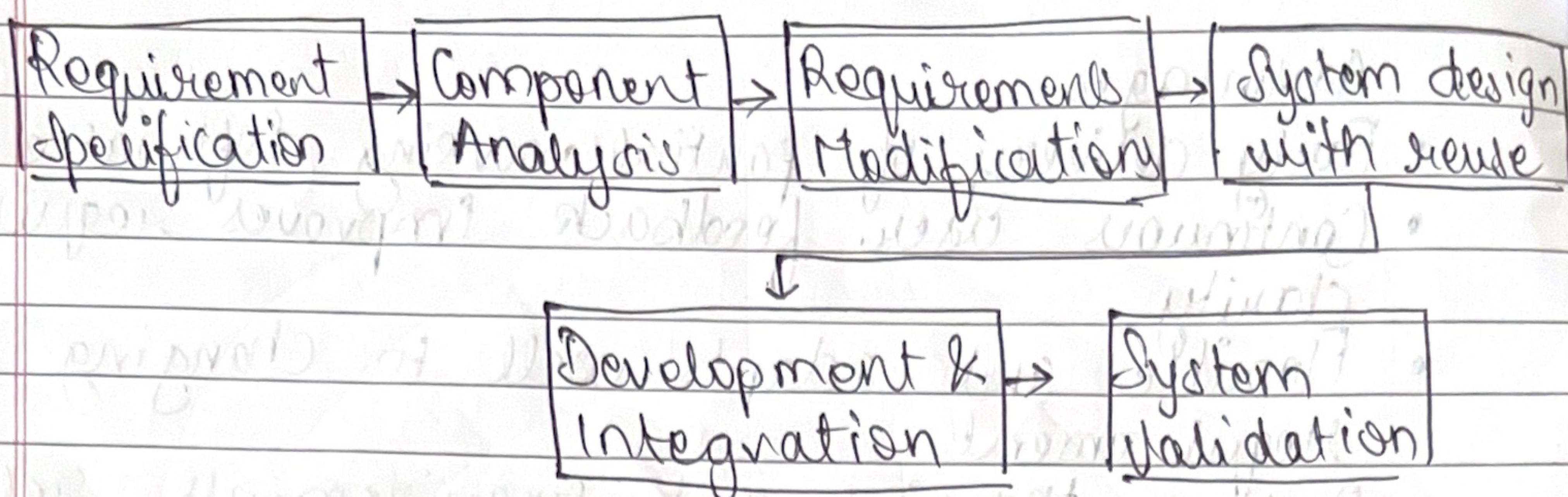
- Difficult to estimate cost and schedule.
- System design may become inconsistent if not well planned.
- Requires constant customer involvement.
- Documentation may be incomplete.

## 3 Component Based Software Engineering Model (CBSE)

### Description

CBSE focuses on building systems by reusing existing software components.

The process includes requirement specification, component analysis, requirement modification, system design with reuse, development, integration and validation.



### Advantages

- Faster development time
- Reduced development and maintenance cost
- High reliability due to tested components
- Encourages software reuse

### Disadvantages

- Limited control over reused components
- Compatibility and integration issues
- Performance may be affected
- Suitable components may not always be available

## 4 Iterative Model

### Description

In the iterative model, the software is developed in small increments called iterations. Each iteration goes through planning, design, coding and testing, producing a working version of the system that improves with each cycle.

### Incremental Model

- Development & delivery in increments

### Spiral Model

- Risk driven processes

### Agile Model

- Focus on rapid delivery of working software

### Advantages

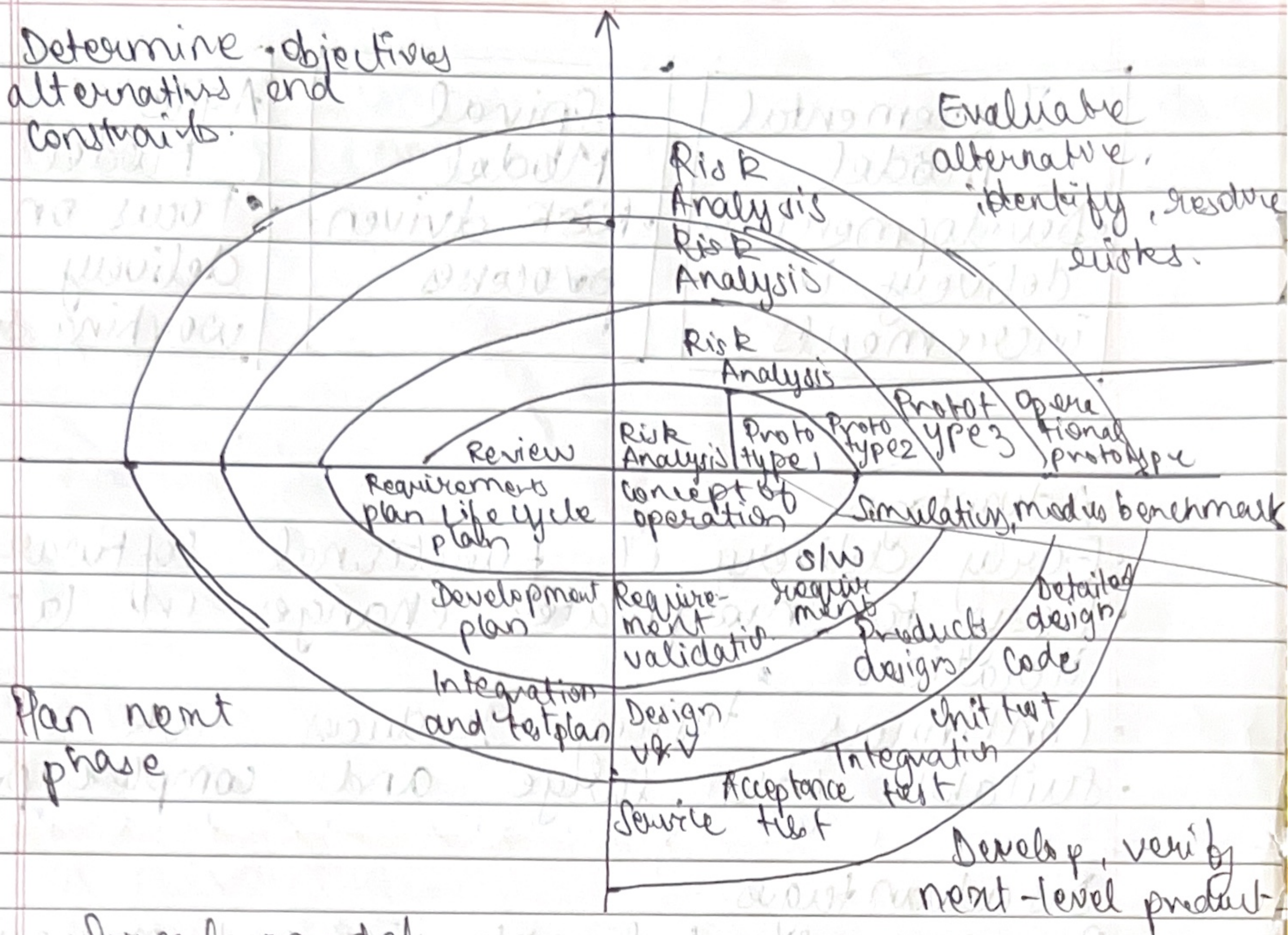
- Early delivery of functional software
- Easy to incorporate changes in later iterations
- Continuous testing reduces risk
- Suitable for large and complex projects.

### Disadvantages

- Requires careful planning and management
- Final cost and timeline are difficult to predict initially.
- Requires skilled developers and active user involvement
- Risk of scope creep.

Determine objectives  
alternatives and  
constraints.

Evaluate  
alternative,  
identify, resolve  
risks.



Plan next  
phase

Spiral model